Namespace Fl.Event.Handling

Classes

ExceptionCatcherEventProcessor<TEvent>

An IEventProcessor<TEvent> decorator that catches any unhandled exceptions thrown by the inner processor and converts them into LanguageExt.Common.Error values.

LoggingEventProcessor<TEvent>

An IEventProcessor<TEvent> decorator that adds structured logging around the inner processor's execution using a Serilog Serilog.ILogger.

ParallelEventProcessor<T>

An IEventProcessor<TEvent> that dispatches an event to all registered IEventHandler<TEvent> instances concurrently.

SequentialEventProcessor<T>

An IEventProcessor<TEvent> that dispatches an event to all registered IEventHandler<TEvent> instances one at a time, in order.

Interfaces

IEventHandler<TEvent>

Defines a handler responsible for processing a single event of type `TEvent`.

IEventProcessor<TEvent>

Defines a processor that orchestrates the handling of events of type `TEvent`.

Class ExceptionCatcherEventProcessor<TEvent>

Namespace: Fl.Event.Handling

Assembly: Fl.Event.Handling.dll

An IEventProcessor<TEvent> decorator that catches any unhandled exceptions thrown by the inner processor and converts them into LanguageExt.Common.Error values.

```
public class ExceptionCatcherEventProcessor<TEvent> : IEventProcessor<TEvent> where
TEvent : class
```

Type Parameters

`TEvent`

The type of the event to process. Must be a reference type.

Inheritance

object ← ExceptionCatcherEventProcessor<TEvent>

Implements

IEventProcessor<TEvent>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ExceptionCatcherEventProcessor(IEventProcessor<TEvent>)

An IEventProcessor<TEvent> decorator that catches any unhandled exceptions thrown by the inner processor and converts them into LanguageExt.Common.Error values.

```
public ExceptionCatcherEventProcessor(IEventProcessor<TEvent> processor)
```

Parameters

`processor` IEventProcessor<TEvent>

The inner IEventProcessor<TEvent> to wrap.

Methods

ProcessAsync(TEvent)

Asynchronously processes the specified event, catching any exceptions thrown by the inner processor and returning them as LanguageExt.Common.Error values.

```
public EitherAsync<Error, Unit> ProcessAsync(TEvent evt)
```

Parameters

`evt` TEvent

The event instance to process.

Returns

EitherAsync<Error, Unit>

An LanguageExt.EitherAsync<L, R> that resolves to LanguageExt.Unit on success, or an LanguageExt.Common.Error wrapping the caught exception if processing fails.

Interface IEventHandler<TEvent>

Namespace: Fl.Event.Handling

Assembly: Fl.Event.Handling.dll

Defines a handler responsible for processing a single event of type `TEvent`.

```
public interface IEventHandler<TEvent> where TEvent : class
```

Type Parameters

`TEvent`

The type of the event to handle. Must be a reference type.

Methods

HandleAsync(TEvent)

Asynchronously handles the specified event.

```
EitherAsync<Error, Unit> HandleAsync(TEvent evt)
```

Parameters

`evt` TEvent

The event instance to handle.

Returns

EitherAsync<Error, Unit>

An LanguageExt.EitherAsync<L, R> that resolves to LanguageExt.Unit on success, or an LanguageExt.Common.Error if handling fails.

Interface IEventProcessor<TEvent>

Namespace: Fl.Event.Handling

Assembly: Fl.Event.Handling.dll

Defines a processor that orchestrates the handling of events of type `TEvent`.

```
public interface IEventProcessor<TEvent> where TEvent : class
```

Type Parameters

`TEvent`

The type of the event to process. Must be a reference type.

Methods

ProcessAsync(TEvent)

Asynchronously processes the specified event.

```
EitherAsync<Error, Unit> ProcessAsync(TEvent evt)
```

Parameters

`evt` TEvent

The event instance to process.

Returns

EitherAsync<Error, Unit>

An LanguageExt.EitherAsync<L, R> that resolves to LanguageExt.Unit on success, or an LanguageExt.Common.Error if processing fails.

Class LoggingEventProcessor<TEvent>

Namespace: Fl.Event.Handling

Assembly: Fl.Event.Handling.dll

An IEventProcessor<TEvent> decorator that adds structured logging around the inner processor's execution using a Serilog Serilog.ILogger.

```
public class LoggingEventProcessor<TEvent> : IEventProcessor<TEvent> where TEvent
: class
```

Type Parameters

`TEvent`

The type of the event to process. Must be a reference type.

Inheritance

object ← LoggingEventProcessor<TEvent>

Implements

IEventProcessor<TEvent>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

LoggingEventProcessor(IEventProcessor<TEvent>, ILogger)

An IEventProcessor<TEvent> decorator that adds structured logging around the inner processor's execution using a Serilog Serilog.ILogger.

```
public LoggingEventProcessor(IEventProcessor<TEvent> processor, ILogger logger)
```

Parameters

`processor` IEventProcessor<TEvent>

The inner IEventProcessor<TEvent> to wrap.

`logger` ILogger

The Serilog logger used to record processing outcomes.

Methods

ProcessAsync(TEvent)

Asynchronously processes the specified event and logs the outcome via the configured Serilog.ILogger.

```
public EitherAsync<Error, Unit> ProcessAsync(TEvent evt)
```

Parameters

`evt` TEvent

The event instance to process.

Returns

EitherAsync<Error, Unit>

An LanguageExt.EitherAsync<L, R> that resolves to LanguageExt.Unit on success, or an LanguageExt.Common.Error if processing fails.

Class ParallelEventProcessor<T>

Namespace: Fl.Event.Handling

Assembly: Fl.Event.Handling.dll

An IEventProcessor<TEvent> that dispatches an event to all registered IEventHandler<TEvent> instances concurrently.

```
public class ParallelEventProcessor<T> : IEventProcessor<T> where T : class
```

Type Parameters

T

The type of the event to process. Must be a reference type.

Inheritance

object ← ParallelEventProcessor<T>

Implements

IEventProcessor<T>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ParallelEventProcessor(IEnumerable<IEventHandler<T>>)

An IEventProcessor<TEvent> that dispatches an event to all registered IEventHandler<TEvent> instances concurrently.

```
public ParallelEventProcessor(IEnumerable<IEventHandler<T>> eventHandlers)
```

Parameters

`eventHandlers` IEnumerable<IEventHandler<T>>

The collection of handlers to invoke in parallel.

Methods

ProcessAsync(T)

Asynchronously dispatches the specified event to all registered handlers in parallel, awaiting all of them before returning.

```
public EitherAsync<Error, Unit> ProcessAsync(T evt)
```

Parameters

`evt` T

The event instance to process.

Returns

EitherAsync<Error, Unit>

An LanguageExt.EitherAsync<L, R> that resolves to LanguageExt.Unit when all handlers succeed, or an LanguageExt.Common.Error if any handler fails.

Class SequentialEventProcessor<T>

Namespace: Fl.Event.Handling

Assembly: Fl.Event.Handling.dll

An IEventProcessor<TEvent> that dispatches an event to all registered IEventHandler<TEvent> instances one at a time, in order.

```
public class SequentialEventProcessor<T> : IEventProcessor<T> where T : class
```

Type Parameters

T

The type of the event to process. Must be a reference type.

Inheritance

object ← SequentialEventProcessor<T>

Implements

IEventProcessor<T>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SequentialEventProcessor(IEnumerable<IEventHandler<T>>)

An IEventProcessor<TEvent> that dispatches an event to all registered IEventHandler<TEvent> instances one at a time, in order.

```
public SequentialEventProcessor(IEnumerable<IEventHandler<T>> eventHandlers)
```

Parameters

`eventHandlers` IEnumerable<IEventHandler<T>>

The ordered collection of handlers to invoke sequentially.

Methods

ProcessAsync(T)

Asynchronously dispatches the specified event to all registered handlers one at a time, stopping early if any handler returns an LanguageExt.Common.Error.

```
public EitherAsync<Error, Unit> ProcessAsync(T evt)
```

Parameters

`evt` T

The event instance to process.

Returns

EitherAsync<Error, Unit>

An LanguageExt.EitherAsync<L, R> that resolves to LanguageExt.Unit when all handlers succeed, or an LanguageExt.Common.Error if any handler fails.

Namespace Fl.Event.Handling.Extensions

Classes

ServiceCollectionExtensions

Provides extension methods for registering event handling services with the IServiceCollection dependency injection container.

Enums

EventHandlingStrategy

Specifies the strategy used to dispatch events to their registered handlers.

Enum EventHandlingStrategy

Namespace: Fl.Event.Handling.Extensions

Assembly: Fl.Event.Handling.Extensions.dll

Specifies the strategy used to dispatch events to their registered handlers.

```
public enum EventHandlingStrategy
```

Fields

`Parallel = 1`

Handlers are invoked concurrently and all are awaited before the result is returned.

`Sequential = 0`

Handlers are invoked one at a time, in registration order. Processing stops immediately if any handler returns an error.

Class ServiceCollectionExtensions

Namespace: Fl.Event.Handling.Extensions

Assembly: Fl.Event.Handling.Extensions.dll

Provides extension methods for registering event handling services with the IServiceCollection dependency injection container.

```
public static class ServiceCollectionExtensions
```

Inheritance

object ← ServiceCollectionExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

AddEventHandling<TEvent>(IServiceCollection, EventHandlingStrategy, Assembly)

Registers an IEventProcessor<TEvent> and all IEventHandler<TEvent> implementations found in the specified `assembly` with the DI container.

```
public static IServiceCollection AddEventHandling<TEvent>(this IServiceCollection
services, EventHandlingStrategy strategy, Assembly assembly) where TEvent : class
```

Parameters

`services` IServiceCollection

The IServiceCollection to add the services to.

`strategy` EventHandlingStrategy

The EventHandlingStrategy that determines whether handlers are invoked sequentially or in parallel.

`assembly` Assembly

The Assembly to scan for IEventHandler<TEvent> implementations.

Returns

IServiceCollection

The original `services` instance for method chaining.

Type Parameters

`TEvent`

The event type to configure handling for. Must be a reference type.

Exceptions

ArgumentException

Thrown when `strategy` is not a recognised EventHandlingStrategy value.